

A State-of-the-Art Review of Cyber Threat Intelligence Awareness Programs in Mitigating Bank Cyber Attacks

Robby Anzil Firdaus
Interdisciplinary School of
Management Technology

Institut Teknologi Sepuluh Nopember
Surabaya
Surabaya, Indonesia
ITSEC Asia
Jakarta, Indonesia
robby@itsecasia.com

Nur Aini Rakhmawati
Information Systems Department
Institut Teknologi Sepuluh Nopember
Surabaya
Surabaya, Indonesia.
nur.aini@its.ac.id

Febriliyan Samopa
Information Systems Department
Institut Teknologi Sepuluh Nopember
Surabaya
Surabaya, Indonesia
iyan@its.ac.id

Abstract— *The importance of cyber threat intelligence (CTI) in combating the increasing threat of cyber-attacks, particularly in the context of bank phishing, cannot be overstated. This study investigates the integration of CTI in bolstering cybersecurity awareness programs within the banking sector, with the aim of identifying effective strategies for safeguarding financial data. The primary objective of this study is to explore how banks develop and implement CTI-driven awareness initiatives, highlighting best practices, challenges, and opportunities. A systematic literature review (SLR) conducted using the PRISMA methodology was employed to synthesize findings from high-quality relevant literature. The findings of this study indicate that CTI can significantly enhance cybersecurity awareness through tactical, operational, and strategic intelligence. Specifically, tactical intelligence involves the collection of indicators of compromise (IoCs), hash values, and malware behaviour descriptions for immediate threat response. Operational intelligence focuses on real-time threat analysis to understand attack mechanisms and improve situational awareness. Strategic intelligence offers long-term security strategies and insights into emerging threats. Integrating a CTI with security information and event management (SIEM) systems enables financial institutions to rapidly identify and address malicious activities, strengthening their cybersecurity defenses to protect their consumers.*

Keywords — *Cyber Threat Intelligence, Phishing Attacks, Cybersecurity Awareness, Banking Sector, Incident Response.*

I. INTRODUCTION

This study explores the critical role of CTI in combating the escalating threat of bank phishing attacks. CTI focuses on researching cyber-attacks, analysing emerging trends, and providing valuable insights to enhance cybersecurity measures [1]. By understanding the developments in cyber threats, the CTI enables security practitioners to recognize attack indicators, extract attack methods, and respond promptly and accurately to mitigate potential risks [2]. This proactive approach is essential in banking, where phishing attacks pose a significant threat to financial institutions and their consumers. Bank-phishing attacks are a prevalent form of cybercrime that target individuals' sensitive financial

information through deceptive means. These attacks often involve fraudulent emails or websites that mimic legitimate banking platforms to trick users into divulging confidential data, such as login credentials or financial details. The CTI plays a crucial role in this landscape by providing valuable insights into the tactics, techniques, and procedures employed by threat actors in executing phishing campaigns [3].

In recent years, the sophistication of phishing attacks has increased, necessitating advanced cybersecurity measures to safeguard against evolving threats. CTI leverages technologies such as artificial intelligence (AI) and machine learning to enhance threat detection capabilities and improve incident response times [4]. The banking sector is a prime target for cybercriminals because of the vast amounts of sensitive financial data it holds. Therefore, financial institutions must adopt robust cybersecurity strategies to protect their assets and customers from phishing attacks. By leveraging CTI frameworks and methodologies, banks can enhance their threat intelligence capabilities, detect malicious activities, and respond swiftly to security incidents [2].

The objective of this study is to investigate and analyse the strategies employed by banks in developing and enhancing awareness programs driven by CTI. This study contributes to the existing body of knowledge by examining how banks strategize in creating and developing awareness programs driven by CTI by synthesizing insights from various disciplines.

II. THEORITICAL CONTEXT

A. Cyber Threat Intelligence

CTI is a critical element in modern cybersecurity practices, offering organizations evidence-based knowledge about potential and existing threats [5]. It encompasses the collection, processing, analysis, and dissemination of contextualized information to comprehend malicious threat actors' motivations, goals, targets, and attack behaviors. By continuously monitoring threats in real-time network

environments, threat intelligence assists in identifying and mitigating security vulnerabilities [6].

Furthermore, the quality of threat intelligence is essential for effective exchange and utilization. When utilized correctly, threat intelligence can support security teams in defending themselves against evolving threats throughout the attack lifecycle. It plays a crucial role in staying ahead of cyber threats by enabling continuous monitoring, risk management, and proactive threat detection [7].

B. Cyber Security Awareness

Cybersecurity awareness is essential for protecting individuals, organizations, and systems from cyber threats. It involves understanding the risks, best practices, and measures to safeguard sensitive information and digital assets from malicious actors. Various theoretical perspectives and research studies have contributed to enhancing cybersecurity awareness. An important aspect is evaluating cybersecurity awareness, which includes assessing the knowledge, attitudes, and behaviors of individuals towards cybersecurity practices [8].

Through systematic reviews of methodologies for assessing cybersecurity awareness, researchers can gain insights into the effectiveness of awareness programs and areas for improvement. These frameworks help create policies and procedures that foster a culture of cybersecurity awareness in educational settings. Additionally, the cost-benefit analysis framework for cybersecurity awareness training programs provides a structured approach to assess the effectiveness and efficiency of training initiatives [9]. By evaluating the costs and benefits of cybersecurity awareness programs, organizations can make informed decisions regarding resource allocation and program optimization. Studies that assess cybersecurity awareness levels among specific groups, such as students or employees, offer valuable insights into the effectiveness of awareness initiatives and the factors influencing awareness levels [10].

C. Mitigating Bank Cyber Attack

Cybersecurity threats pose significant challenges to the banking sector and require robust mitigation strategies. This literature review focuses on the effectiveness of cybersecurity awareness programs in reducing phishing and malware attacks on large banks. A holistic approach to cybersecurity, emphasizing collaboration, innovation, and education, has been advocated as essential for building a secure banking ecosystem, as demonstrated in a case study of Indian banks [11]. By leveraging the CTI, security teams can gain a comprehensive understanding of adversary strategies and enhance defensive measures, contributing to proactive defense mechanisms [12].

Traditional cyber threats, including phishing and malware attacks, have been a persistent challenge in the banking sector, necessitating robust risk assessment methodologies and best practices for effective cybersecurity defense. Understanding the evolving landscape of phishing methods and implementing preventive measures are crucial for mitigating these threats [13].

By examining real-world incidents and case studies, insights into the dynamic nature of cyber threats are gained, underscoring the importance of proactive and adaptive cybersecurity measures in the banking sector. Furthermore, the integration of SIEM, SOAR, and AI technologies enables

cybersecurity teams to anticipate and respond to emerging threats effectively, thereby enhancing cyber resilience [14].

III. METHODS

Implementing a CTI program can lead to substantial improvements in cybersecurity metrics. Studies have shown that organizations that adopt CTI witness an increase in threat detection and a decrease in security incidents compared to those without CTI. The CTI aids in understanding the network landscape and enables proactive measures against potential threats [15].

CTI plays a crucial role in strengthening traditional cybersecurity strategies by providing indicators of compromise (IoCs) derived from recent cyber threats, thereby enhancing an organization's ability to mitigate attacks more effectively and efficiently. By generating timely, relevant, and actionable intelligence on emerging threats, the CTI ensures that organizations can respond swiftly to incidents, thereby improving response times [16].

In accordance with the methodology outlined by Kitchenham [17], this study was conducted in three stages: planning, conducting, and reporting. The planning phase involved formulating the research questions, defining the search string, selecting publication sources, and establishing the inclusion and exclusion criteria. The systematic literature review (SLR) followed the PRISMA guidelines to ensure a structured and transparent review process [18].

A. Defining the Search String

A comprehensive search string was developed by incorporating relevant keywords and Boolean operators to retrieve studies related to CTI, phishing attacks, cybersecurity awareness, and the banking sector. As shown in Table I, the search string was iteratively refined to enhance the precision and recall of the search results [19].

TABLE I. SEARCH STRINGS USED IN THE STUDY

Code	Formula used in this study for search strings
SS-1	("cyber threat intelligence" OR "CTI" OR "threat intelligence")
SS-2	("awareness program" OR "training" OR "education" OR "simulation")
SS-3	("utilization" OR "use" OR "implementation" OR "adoption")
SS	SS-1 AND SS-2 AND SS-3

B. Publication Sources for Search

Publication sources were selected based on their relevance to the research topic and the likelihood of containing valuable insights into banks' cybersecurity awareness initiatives. Databases such as Emerald, Science direct, Taylor & Francis and IEEE were included to ensure a broad coverage of the literature

Emerald	= 292 paper
IEEE	= 304 paper
Science direct	= 291 paper
Taylor & Francis	= 61 paper
Total of paper	= 948 paper

C. Inclusion and Exclusion Criteria

As shown in Table II, clear inclusion and exclusion criteria were established to determine the eligibility of the studies for the review. Criteria related to publication date, study design, relevance to research questions, and language were defined to ensure the selection of high-quality and pertinent literature [20].

TABLE II. INCLUSION AND EXCLUSION FORMULA

Category	Include	Exclude
Population	Banks or financial institutions.	Non-financial organizations.
	Studies involving bank employees, customers, or security personnel.	Studies not related to banks or financial institutions.
Intervention / Exposure	CTI strategies.	General cybersecurity strategies not related to threat intelligence.
	Awareness programs designed to mitigate cyber threats.	Awareness programs not related to cyber threats.
Comparator / Context	Comparison of different CTI strategies.	Comparisons not involving CTI.
	Contexts involving phishing attacks and cyber threats in banking.	Contexts outside of banking or financial sectors.
Outcome	Effectiveness of awareness programs.	Outcomes not related to awareness programs or phishing attacks.
	Reduction in attack incidents.	Outcomes not measured in the context of banks or financial institutions.
Study Characteristics	Peer-reviewed articles.	Non-peer-reviewed articles.
	Case studies.	Editorials and opinion pieces.
Time	Studies published within the last 10 years.	Studies older than 10 years.
	Studies in English or Indonesian.	Studies not in English or Indonesian.

A systematic literature review was conducted following the PRISMA guidelines, which outline a structured approach to screening, data extraction, and quality assessment of the selected studies [21]. The inclusion and exclusion formula in Table 2 was used as a criterion in the Covidence app, a web-based platform for systematic reviews, to streamline the screening process, facilitate collaboration among reviewers, and ensure consistency in study selection [18].

IV. RESULTS AND DISCUSSIONS

A. Identification Phase

The process starts with the identification phase shown in Fig. 1. in which a comprehensive search is conducted across various databases and registers using specified search keywords. This search yielded 948 studies. During this phase, 52 references were identified as duplicates.

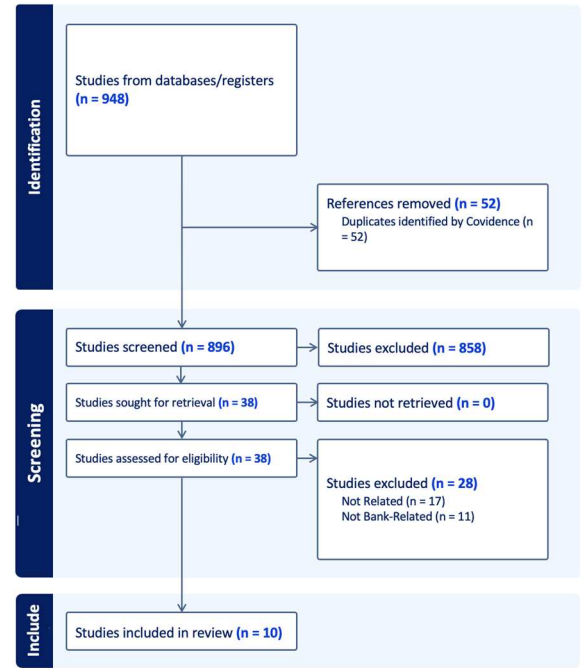


Fig. 1. Article screening (based on the PRISMA flow diagram)

B. Screening Phase

During the screening phase, as shown in Fig. 1, the remaining 896 studies underwent a preliminary evaluation based on their titles and abstracts. A total of 858 studies were excluded from the initial screening. Only 38 studies were included for full-text retrieval.

The eligibility phase in Fig. 1 involved a detailed examination of the full text of the 38 retrieved studies. Each study was scrutinized to ensure that it was relevant, methodologically sound, and provided sufficient data to address the research question. During this phase, 28 studies were excluded for specific reasons: 17 studies were unrelated to the research question. In addition, 11 studies were not specifically related to banks.

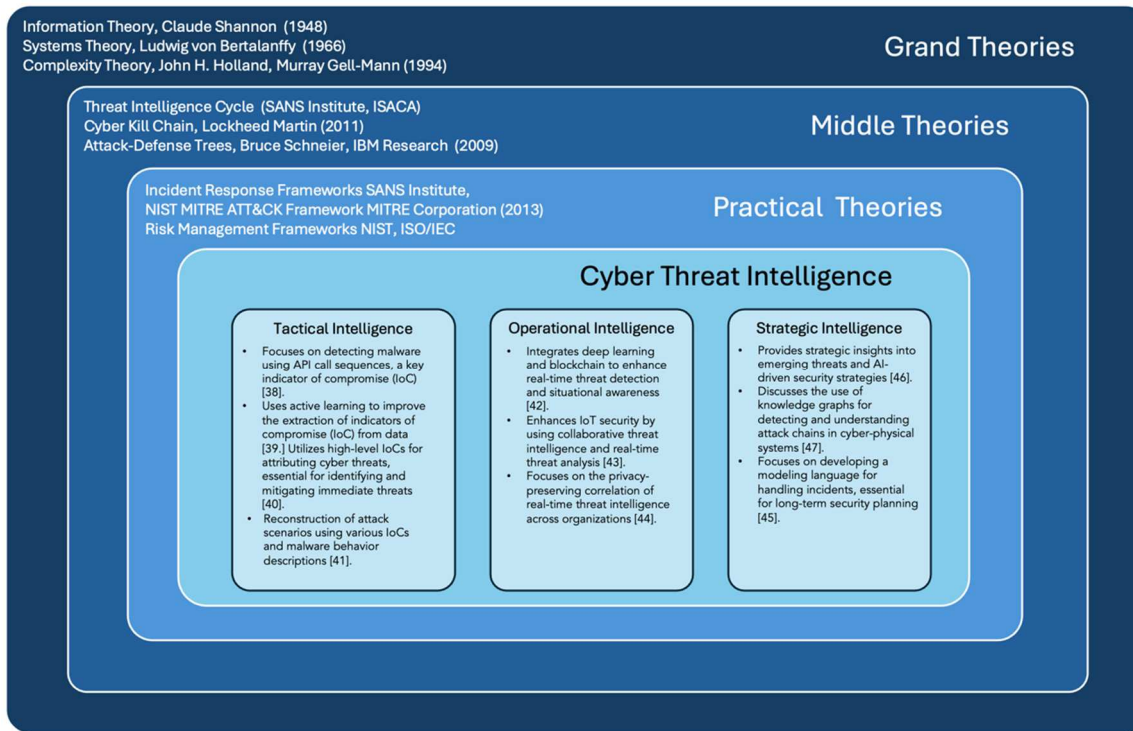


Fig. 2. Scheme of Cyber Threat Intelligence

C. Include Phase

Finally, as seen in Fig. 1, in the included phase, 10 studies met all the inclusion criteria and were included in the systematic review. These studies represent the final set of data sources that will be analysed to answer the research question. They provide detailed information on how banks develop and implement awareness programs using CTI, offering valuable insights and evidence to support the conclusions of this review

D. Data Analysing

In the context of analyzing a small quantity of data, such as the 10 papers resulting from screening, as shown in Fig. 1. This suggests that a standard word-processing package is sufficient for the analysis process. In addition, Carvalho and Santos [23]. pointed out that content analysis, although recognized for its appropriateness in qualitative studies, may face challenges related to the subjectivity of the coding process, potentially affecting the assurance of validity criteria. This implies that manual analysis methods can still be valuable, even in the absence of extensive data processing software. Future investigations. Tong [24] emphasized that synthesizing qualitative research can lead to the development of new theoretical models, identification of research gaps, and support for the creation and evaluation of health interventions. Brown [25] further highlighted that data synthesis enables the combination of diverse data sources and expert knowledge to produce new information that aids decision-making processes. These references underscore the importance of data synthesis in generating a comprehensive understanding and facilitating evidence-based decision-making.

E. Cyber Threat Intelligence Driven Awareness

Focusing on the three main types of analysis in CTI—Tactical, Operational, and Strategic Intelligence—it becomes evident that these forms of intelligence are crucial in enhancing cybersecurity preparedness and response in the banking sector. Tactical Intelligence provides essential Indicator of Compromise (IoC) data for rapid response, Operational Intelligence offers real-time analysis to strengthen situational awareness, and Strategic Intelligence provides long-term insights aiding in security strategy development. Studies have shown that these three types of intelligence have a significant impact on improving cybersecurity metrics within the banking sector, outperforming other forms of analysis in terms of effectiveness [26].

Tactical Intelligence, by focusing on IoCs, enables banks to respond swiftly to potential threats, enhancing their ability to detect and mitigate security incidents promptly. Operational Intelligence, with its real-time analysis capabilities, ensures that banks have a comprehensive understanding of their cybersecurity landscape, allowing for proactive threat management and incident response. Strategic Intelligence, offering insights for long-term security planning, and assists banks in developing robust security strategies that align with their overall business objectives, thereby enhancing their resilience against evolving cyber threats. By prioritizing these three key forms of intelligence, banks can establish a comprehensive cybersecurity framework that addresses threats at various levels, from immediate responses to long-term strategic planning. [16].

The theoretical foundation of CTI is built upon grand, middle-range, and practical theories, as seen in Fig 2. Each contributes uniquely to our understanding and application of the CTI. Grand Theories provide broad, abstract frameworks essential for grasping large-scale phenomena within the CTI.

Claude Shannon's Information Theory lays the foundational principles for data transmission and security, crucial for understanding how information about cyber threats is communicated and processed efficiently [27]. Ludwig von Bertalanffy's Systems Theory introduces a perspective where organizations are viewed as complex, interrelated systems, aiding in comprehending the intricate interactions within an organization's cybersecurity infrastructure [28]. Complexity Theory, as pioneered by John H. Holland and Murray Gell-Mann offered insights into the dynamic and adaptive nature of cyber threats and defenses, highlighting the emergent behaviors from interactions within the cybersecurity ecosystem [29].

As seen in Fig 2. Middle-Range Theories bridge the gap between grand theories and practical applications by providing more focused frameworks that address specific aspects of CTI. The Threat Intelligence Cycle, developed by various organizations such as the SANS Institute and ISACA, outlines the structured process of intelligence gathering, including direction, collection, processing, analysis, dissemination, and feedback [30]. This cycle ensures a systematic approach to handling threat intelligence. The Cyber Kill Chain model dissects the stages of a cyberattack from reconnaissance to exfiltration, offering a strategic model to identify and disrupt threats at various stages [31]. Attack-Defense Trees serve as graphical representations to map potential attack vectors and corresponding defenses, providing a visual tool to plan and implement security measures [32]. Practical Theories focus on actionable frameworks and guidelines that can be directly applied in real-world scenarios to enhance cybersecurity [32].

Incident Response Frameworks, developed by the SANS Institute and NIST, provide detailed guidelines for effectively responding to cyber incidents, ensuring that organizations can manage and mitigate the impact of cyber threats efficiently [33]. The MITRE ATT&CK Framework offers a comprehensive knowledge base of adversary tactics and techniques derived from real-world observations, assisting organizations in developing robust defense strategies [34]. Risk Management Frameworks (NIST, ISO/IEC) encompass comprehensive guidelines for managing cybersecurity risks, enabling systematic identification, assessment, and mitigation of risks [35], [36].

Within the realm of CTI, Tactical Intelligence involves the immediate response to threats by gathering indicators of compromise (IoCs), hash values, and descriptions of malware behavior. Chen [37] focused on detecting malware using API call sequences, a critical IoC but their approach may struggle with less predictable behaviours and may not generalize well across different environments. Tang [38] improves the extraction of IoCs using active learning, yet require extensive training data and may not perform well with unstructured data. Noor utilizes high-level IoCs for threat attribution but faces scalability issues and difficulties with novel threats. Noor [39] utilized high-level IoCs for attributing cyber threats, but faced scalability issues and difficulties with novel threats. Zang [40] discussed the reconstruction of attack scenarios using various IoCs and malware behavior descriptions; however, they require substantial resources and may overlook some threats.

Operational Intelligence emphasizes real-time threat analysis to understand attack mechanisms and situations. J S

[41] integrated deep learning and blockchain to enhance real-time threat detection and situational awareness, although this approach is resource-intensive. Nazir [42] leveraged collaborative threat intelligence and real-time analysis to bolster IoT security, which may not always be feasible. Jia [43] focused on the privacy-preserving correlation of cross-organizational threat intelligence, but face challenges where real-time data sharing is limited.

Strategic Intelligence entails long-term security strategies and insights into emerging threats. Abbasizadeh [44] provided strategic insights into emerging threats and AI-driven security strategies, but faced slow adoption and resistance to change. Choo [45] explored the use of knowledge graphs for detecting and understanding attack chains in cyber-physical systems, but may not address immediate security needs and require continuous updates. Casey [46] focused on developing a modeling language for incident handling, which is essential for long-term security planning, which are resource-intensive to implement and may be limited by data integration feasibility.

Overall, each study contributes valuable insights into CTI; however, practical implementation, scalability, and adaptability remain common challenges.

V. CONCLUSION

In conclusion, this study explored the critical role of CTI in enhancing cybersecurity awareness programs within the banking sector. How CTI can significantly enhance cybersecurity awareness, it is evident that CTI plays a pivotal role in improving key cybersecurity metrics such as incident response speed, reduction in security incidents, and increased employee awareness of cyber threats. By synthesizing findings from various reputable sources, we can draw a comprehensive conclusion.

CTI has been shown to enhance organizational cybersecurity resilience by providing timely and actionable intelligence on potential risks within the cyber domain [26]. This intelligence enables organizations to understand and respond effectively to emerging cyber threats, thereby improving the speed of incident responses [16]. Additionally, the CTI aids in the detection of threats, leading to a decrease in the overall number of security incidents experienced by organizations.

Moreover, CTI contributes to raising cybersecurity awareness among employees by providing insights into evolving cyber threats and necessary defense strategies. By fostering a culture of vigilance and knowledge among employees, organizations can empower their workforce to promptly identify and report security incidents, further contributing to a decrease in security breaches [26]. The most pressing area for future research in enhancing cybersecurity within the banking sector through CTI lies in the integration of advanced threat detection tools. This research should investigate the effectiveness of incorporating cutting-edge technologies, such as machine learning and artificial intelligence, with existing CTI frameworks.

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